FraudGuard

Financial Fraud Detection & Customer Risk Analysis

Data Warehouse and Mining (DWM)
Department of INFT
Vidyalankar Institute of Technology
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5,000	25	103	5	3
loan applications	attributes	fraud cases (2.06%)	ML models compared	role-based modules

1 Problem

Fraudulent loan applications are rare (2.06% here) and easily missed. Manual review is subjective, records no reason, and gives no analytical view of fraud.

2 Objective

- Star schema and OLAP analysis of loan data
- Evaluate classifiers on imbalanced data
- Loan workflow with an audit trail
- Role-based access for every module

3 Dataset

5,000 labelled applications, 5 June 2022 to 4 June 2025. 0 missing values, 0 duplicates. Target: **FraudFlag**. No credit score in the data. Data quality: 5 of 8 checks Healthy.

Fraud % by loan type

Business Loan	2.30%
Education Loan	2.14%
Personal Loan	2.14%
Home Loan	1.94%
Car Loan	1.78%

4 Methodology

1. Extract from source, cache as CSV
2. Clean, convert types, derive DTI
3. Impute, scale, one-hot encode (32 features)
4. Stratified 80 / 20 split (4,000 / 1,000)
5. SMOTE on the training fold only
6. Train 5 models in identical pipelines
7. Evaluate on the untouched test set

5 ML Models

Model	Acc.	Prec.	Recall	ROC	Found
Logistic Reg.	60.6%	2.1%	**38.1%**	0.507	8/21
Decision Tree	85.1%	0.8%	4.8%	0.441	1/21
Naive Bayes	66.4%	**2.1%**	33.3%	0.519	7/21
Random Forest	**97.9%**	0.0%	0.0%	0.535	0/21
XGBoost	94.7%	0.0%	0.0%	**0.555**	0/21

Test set: 1,000 applications, 21 fraud. Purple = highest per metric. Found = fraud detected.

6 Key Results

97.9% highest accuracy (Random Forest), but 0% recall

38.1% highest recall (Logistic Regression), 381 false alarms

0.555 highest ROC-AUC (XGBoost); all near 0.5

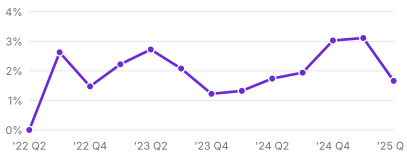
No model detects fraud reliably with these attributes; accuracy alone is misleading.

7 Fraud Detection

1. Model returns a fraud probability
2. Risk: Low < 25%, Medium 25–50%, High ≥ 50%
3. Officer approves, rejects or sends for review
4. Decision and reason kept in audit trail

8 Fraud Trend

Fraud % by quarter of application date



9 DWM / OLAP

Star schema: one fact table, five dimensions

- **Date** — Year · Quarter · Month
- **Loan Application** — FACT TABLE · Amount · Fraud Flag · Income · EMIs · DTI
- **Loan** — Type · Purpose
- **Customer** — Gender · Age Group
- **Employment** — Status
- **Fraud Status** — Flag · Type

Year (roll-up)	Loans	Fraud	Fraud %
2022	932	17	1.82%
2023	1,652	34	2.06%
2024	1,664	33	1.98%
2025	752	19	2.53%

- Drill-down: Year → Quarter → Month → Loan Type
- Slice, dice and pivot on Time, Loan Type, Employment, Gender, Age Group
- Measures: loan count, fraud count, fraud %, loan amount

10 System Architecture

Customer portal · **Bank Officer** workspace · **Admin** technical panel

▼ React + TypeScript

FastAPI backend
Authentication · role-based access · workflow · analytics · OLAP · prediction

▼

Dataset SheetDB / CSV · **SQLite** users, audit trail · **Models** .joblib pipelines

21 API endpoints. Every request is checked against the user role: Customer, Bank Officer or Admin.

11 Modules

- **Customer**: apply for a loan, track status, timeline and rejection reason
- **Bank Officer**: review with fraud probability; approve, reject or send for review
- **Admin / Technical**: dataset, data quality, model comparison, OLAP, system information

12 Conclusion

- Loan data modelled as a star schema and analysed with OLAP
- Five models compared; none detects fraud reliably with these attributes
- Working Customer, Bank Officer and Admin modules with audit trail

13 Future Scope

- Richer features: credit history, repayment behaviour
- Threshold tuning, cross-validation, SHAP explanations
- Star schema in a relational warehouse with scheduled ETL

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